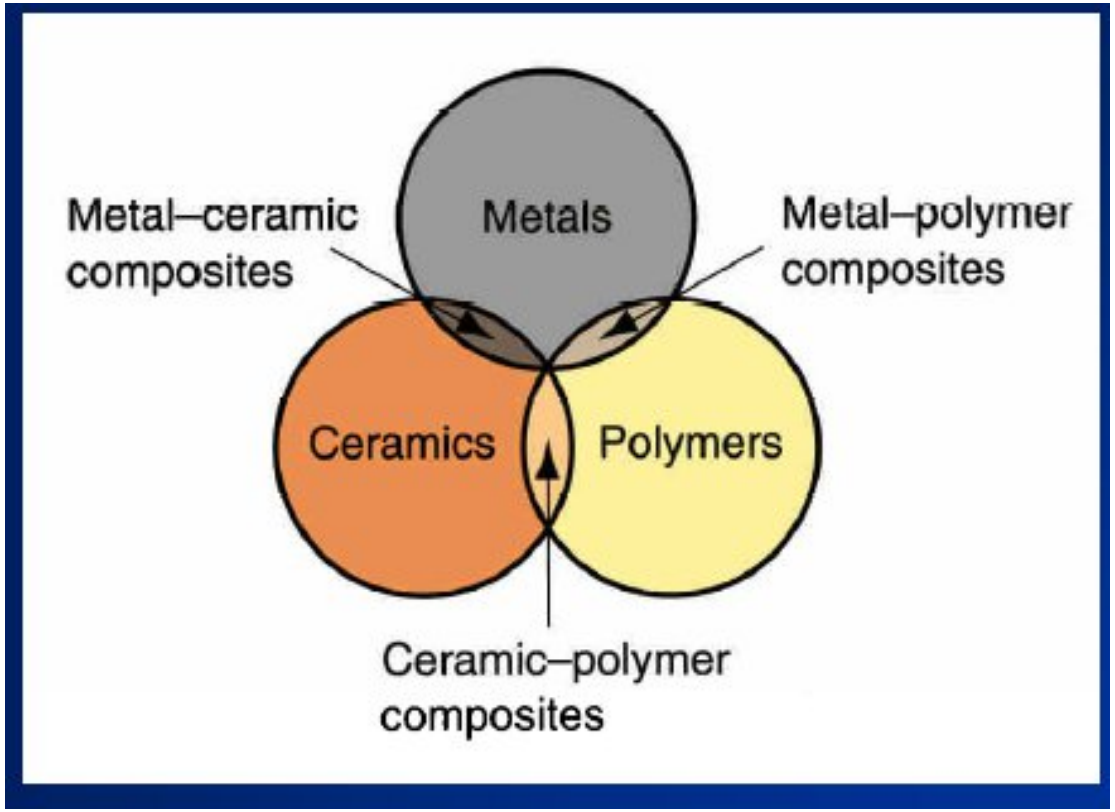
Materials

- On earth the devices like engines, pumps, pressure cookers, cooking vessels, electric motors, bicycle etc. are not readily available.
- They are to be made from materials available on earth.
- Materials and manufacturing helps to realize the conceptual design of engineers.
- Materials are available in nature either as minerals or in raw form.
- They are converted to preform by material processing companies
- Conversion of preformed materials available in the market place into devices and implements is the task of mechanical or production engineers.


Metal–ceramic composites
Metals
Metal–polymer composites
Ceramics
Polymers
Ceramic–polymer composites

FERROUS METALS

Ferrous metals are wrought iron (with less than 0.2% carbon), cast iron (with 2–4.5% carbon), steel (upto 2% carbon). There may also be alloys of iron and other elements. Ferrous alloys are the most commonly used engineering materials. Its density is around $7800\ kg/m^3$.

- **Cast Iron:** Carbon is present in Cast Iron in the free form and it is usually present in the range of 2% to 4.5%. In addition, silicon, manganese, sulfur and phosphorus are also contained in varying amounts. Depending upon the form of carbon present, several types of cast iron are obtained. These include

1. Gray Cast Iron (Due to the gray color of the fracture)
2. White Cast Iron
3. Ductile Cast Iron (Nodular Cast Iron)
4. Malleable Cast Iron

- **Carbon Steels:** Plain carbon Steels are alloys of Iron and Carbon. They are further classified as

1. Low carbon steel (up to 0.30% C)
2. Medium carbon steel (0.30 to 0.60% C)
3. High carbon steel (0.60% to 0.99% C)

- **Alloy Steels:** Steels that contain nickel, chromium, cobalt, vanadium etc. apart from carbon as alloying elements are called as alloy steels.

- **Stainless Steels:** It contains at least 10% chromium, with or without other elements. Stainless steel is used where both the properties of steel and resistance to corrosion are required.

Based on the structures, stainless steels can be grouped into three grades:

NONFERROUS METALS

Nonferrous metals and alloys cover a wide range of materials, from the more common metals such as aluminum, copper and magnesium to high-strength, high-temperature alloys such as tungsten, titanium, tantalum and molybdenum, etc.

Copper

Good corrosion resistance

Good strength

High heat & electrical conductivity

Easily welded, soldered or brazed

Aluminum

Light weight, 1/3 of iron

Weak, ductile and non-magnetic

High thermal & electrical conductivity

Good radiator of energy & reflector of light

Lead (Liquid metal, due to low melting point)

Soft and weak

Malleable and ductile

Heavy and easily meltable (Melting pt: 327.5℃)

Corrosion resistant

Easily Cast, rolled and extruded

- **Aluminum and its alloys:** The main reason in selecting aluminum and its alloys are their high strength-to-weight ratio, their resistance to corrosion for many chemicals, their high thermal and electrical conductivity, reflectivity and appearance and their ease of formability and machinability. Its density is around $2700\ kg/m^3$.

1. **Brass**, which is an alloy of Copper and Zinc and has numerous applications, including decorative objects. The proportions of zinc and copper is varied to create a range of brasses with varying properties.

2. **Bronze**, which is an alloy of copper and tin (modern bronze - 88% copper and 12% tin; architectural bronze - 57% copper, 3% lead, 40% zinc) having good strength and hardness for applications such as bearings, sculptures, etc.

- **Titanium and its alloys:** Titanium and its alloys have excellent corrosion resistance for applications where strength considerations are secondary (Aircrafts, Ships, etc.). Aluminium, vanadium, molybdenum, manganese and other alloying elements are added to titanium alloys to impart properties such as improved workability, strength and hardenability.

- **Magnesium and its alloys:** Magnesium is the lightest engineering metal (1740 kg/m^3) and it has good vibration damping characteristics. Its alloys are used in structural and non-structural applications wherever weight is of primary importance. Magnesium alloys are used in aerospace applications, high-speed machinery, and transportation and materials handling equipment.

- **Plastics:** Plastics are materials that are composed principally of naturally occurring and modified or artificially made polymers often containing additives such as fibers, filler (Fillers are particles added to material like plastics, concrete, etc. to lower the consumption of more expensive binder material or to better some properties. Ex: Calcium carbonate)

Plastics include thermoplastics, thermoset materials and elastomers.

- **Ceramics:** A ceramic is broadly defined as any inorganic non-metallic material. Some of the useful properties of ceramics and glasses are high melting temperature, strength, stiffness, hardness, wear resistance and corrosion resistance and low density.

- **Composites:** Composites are formed from two or more types of materials. Examples include polymer/ceramic and metal/ceramic composites. Since the overall properties of the composites in the required orientations are superior to those of the individual constituents, they are widely used for many engineering applications.

  Ex: Concrete - consists of loose stones (aggregate) held with a matrix of cement.

Stainless steel



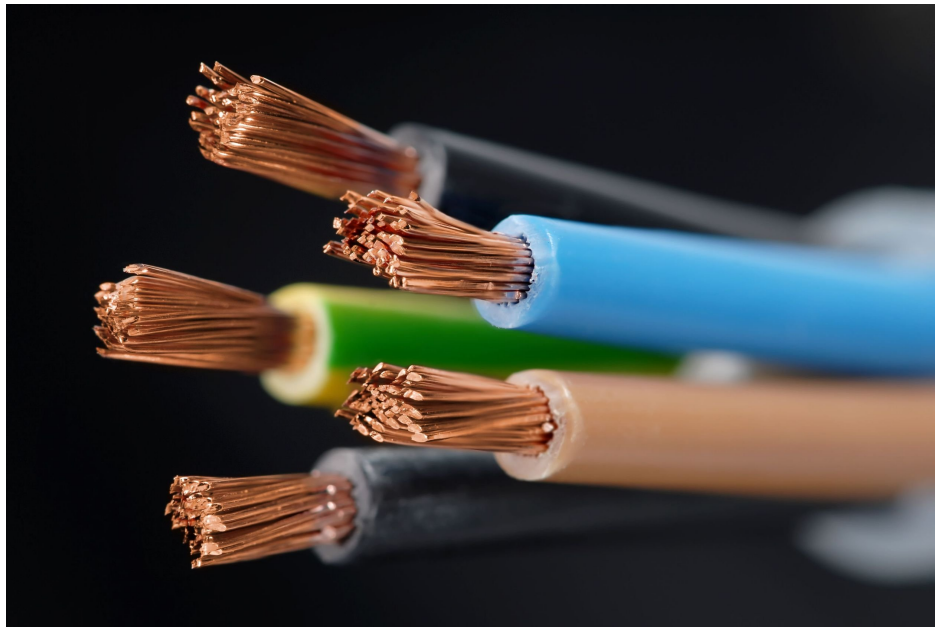




- Corrosion resistance
- Heat conduction average

copper





- Heat conduction
- Electrical conduction
- Prone to oxidation

Brass vessels, lamps, taps

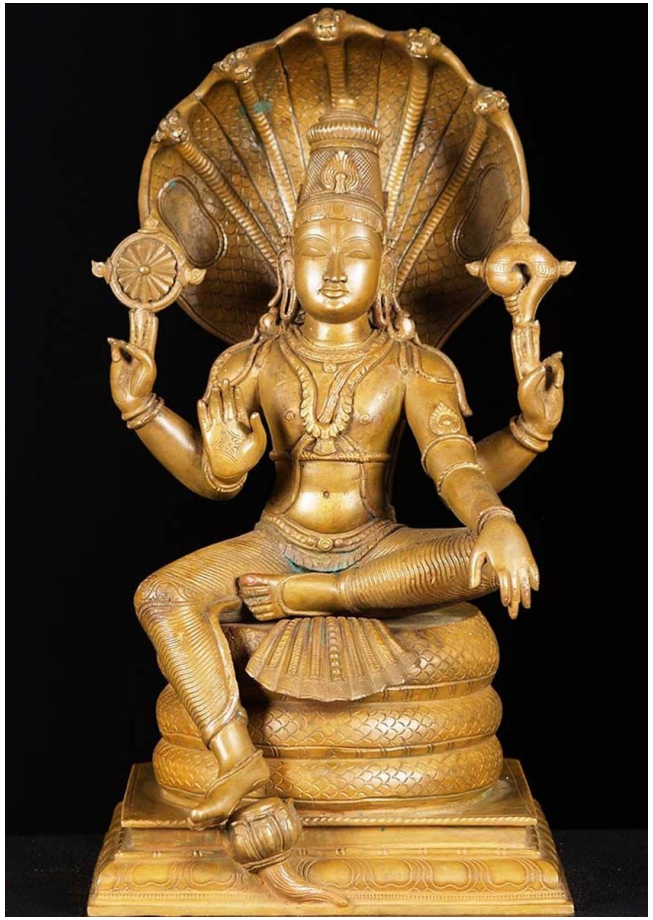






- Corrosion resistance
- appearance

bronze







- Ringing property
- Corrosion resistance

steel








SMITHEY
IRONWARE
DUTCH OVEN
US MADE
SMITHEY
SMITHEY

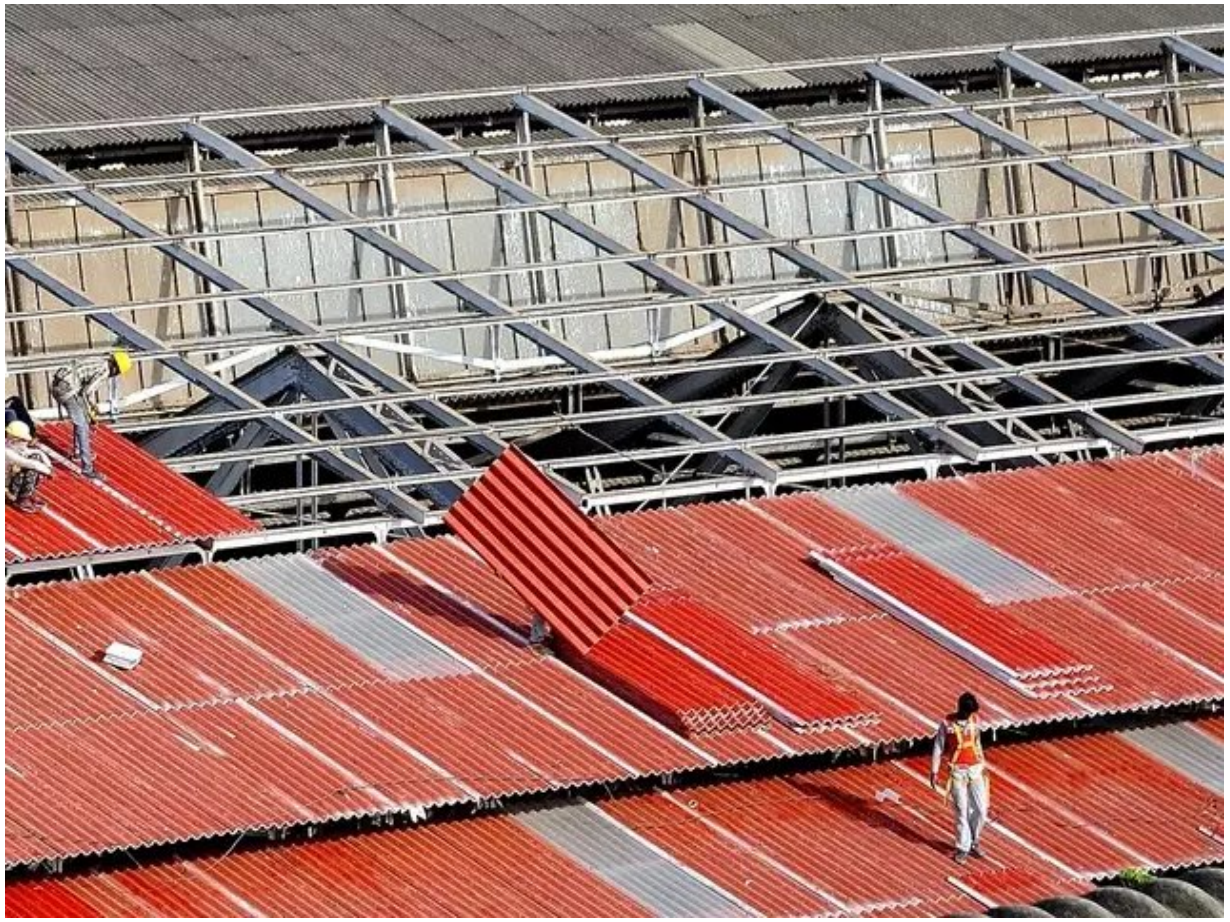




Steel bridges



Steel roof sheets





Steel cans



- Strong
- heavy
- Cheap
- Prone to corrosion
- recyclable

Aluminium






pepsi
Coca-Cola
pepsi
light
Fanta
Sprite
Coca-Cola
zero
7UP
MIRINDA





- Light
- Corrosion resistance

stones





- .





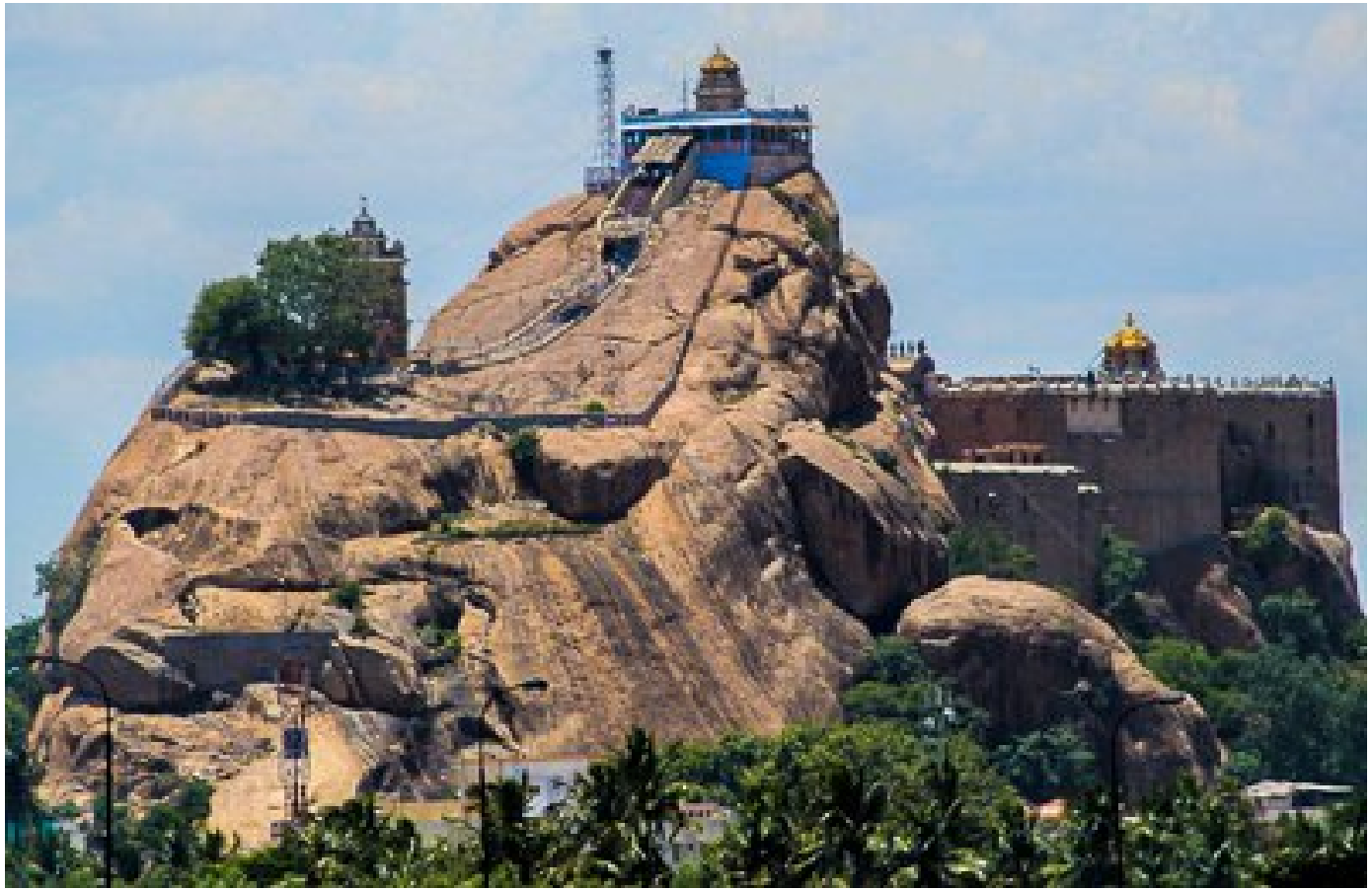
Stone gravel







Rockfort temple Tiruchy



- Brittle
- Cheap
- Heavy
- Long lasting

clay








MCA

- Good insulator of heat
- brittle

glass






alamy stock photo
DDKWPH
www.alamy.com

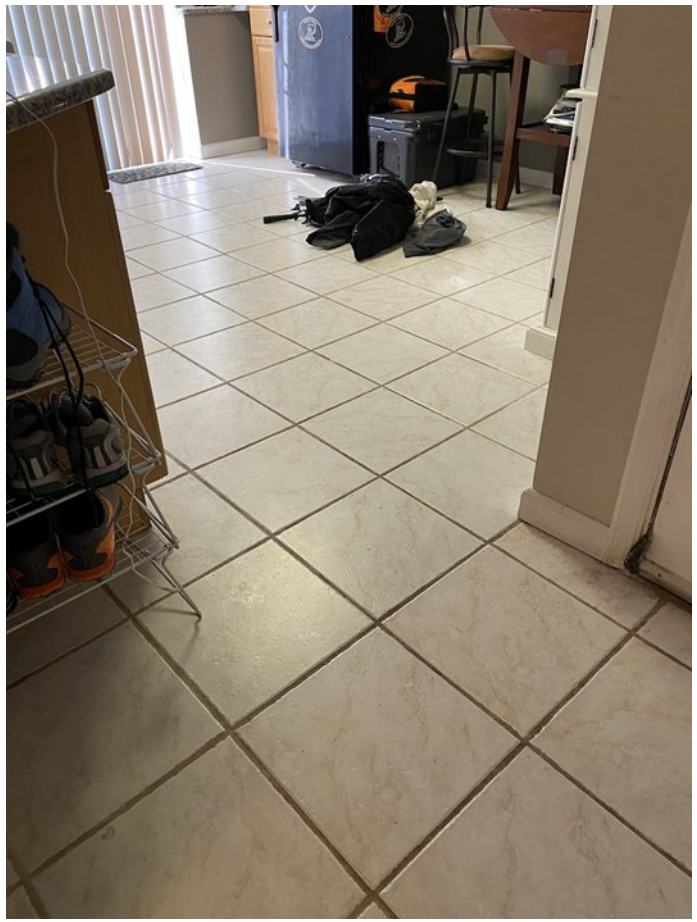


- Transparent
- Brittle
- Inert

porcelain













- Resistance to acid cleaning
- Vitreous- impermeable
- insulator

wood






HOUSE IDEAS


#Funny Life





- Insulator
- Easily workable

concrete






NAVAYUGA
BRAHMAPUTRA RIVER BRIDGE
DHOLA-SADIA

Grand anicut 2000 year old barrage





- Strong
- Easily workable
- Corrosion resistant

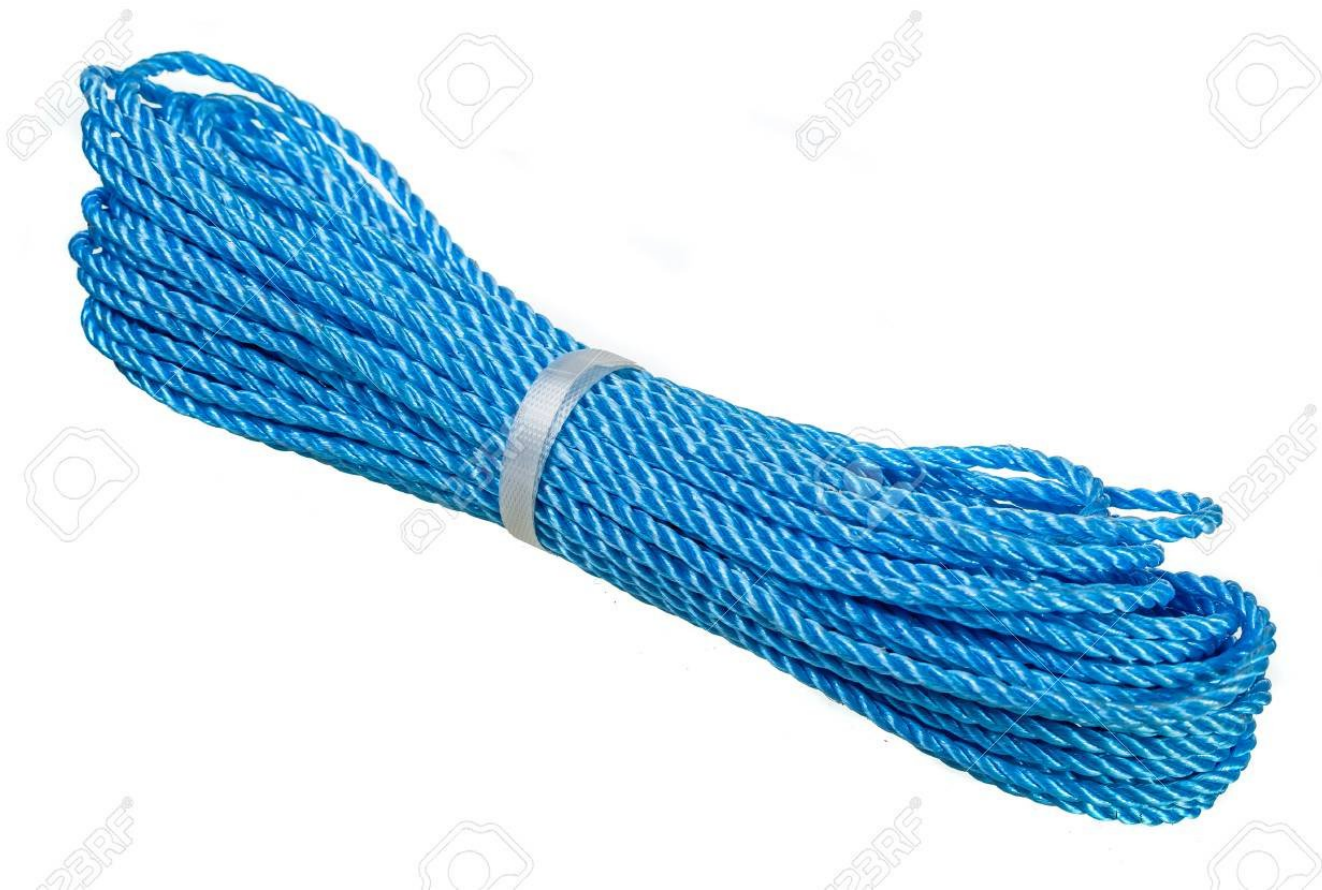
plastic






Optiflesh


Colgate
Fluoride Toothpaste
Whitening
OXYGEN BUBBLES
Brisk Mint Paste







- Cheap
- Easily workable
- Electrical insulator
- Non disposable

rubber








KitchenCraft


Home-Complete






CONVERSE
ALL STAR


WILDPEAK A/T

- Flexible
- Electrical insulator

coir









TEMPLE CAR ROPE (Coir THER VADAM) srirangam







Jute


Jute plants
Harvested plants immersed in water
Rotten stems are made into jute fibre
studyandscore.com

jute







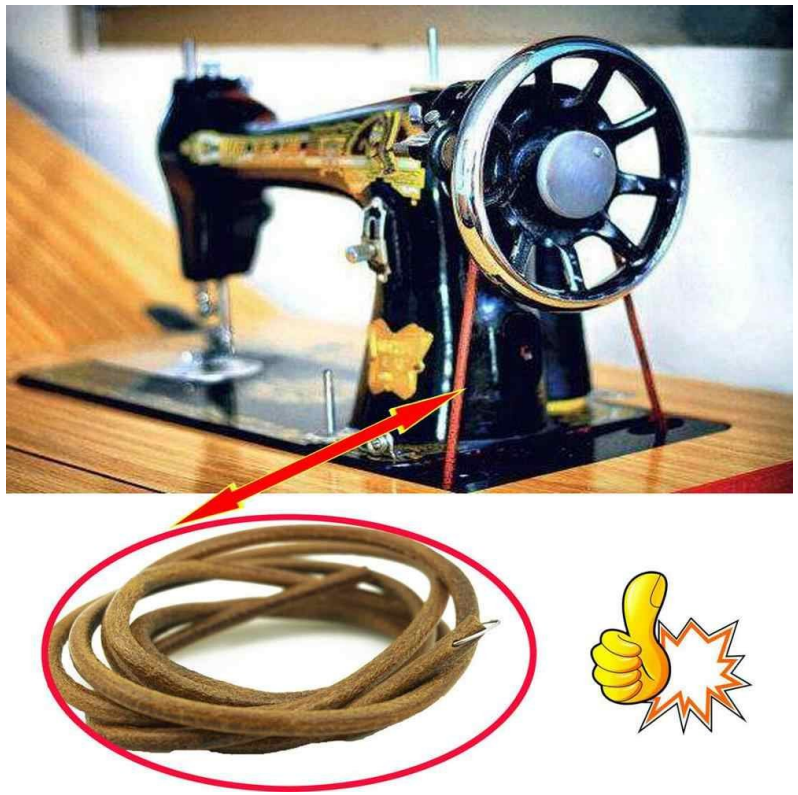


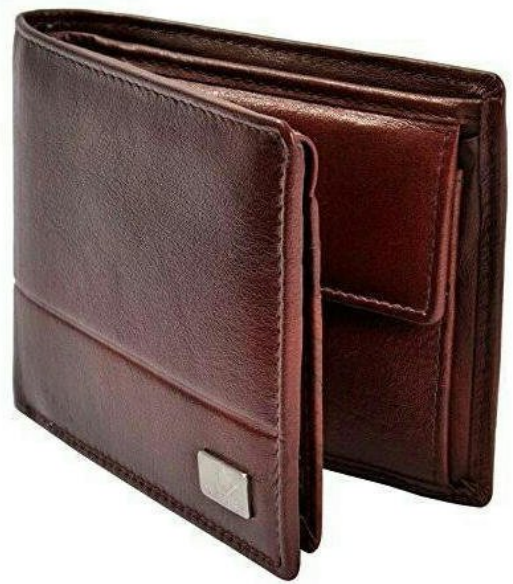
- Strong
- Cheap
- biodegradable

leather









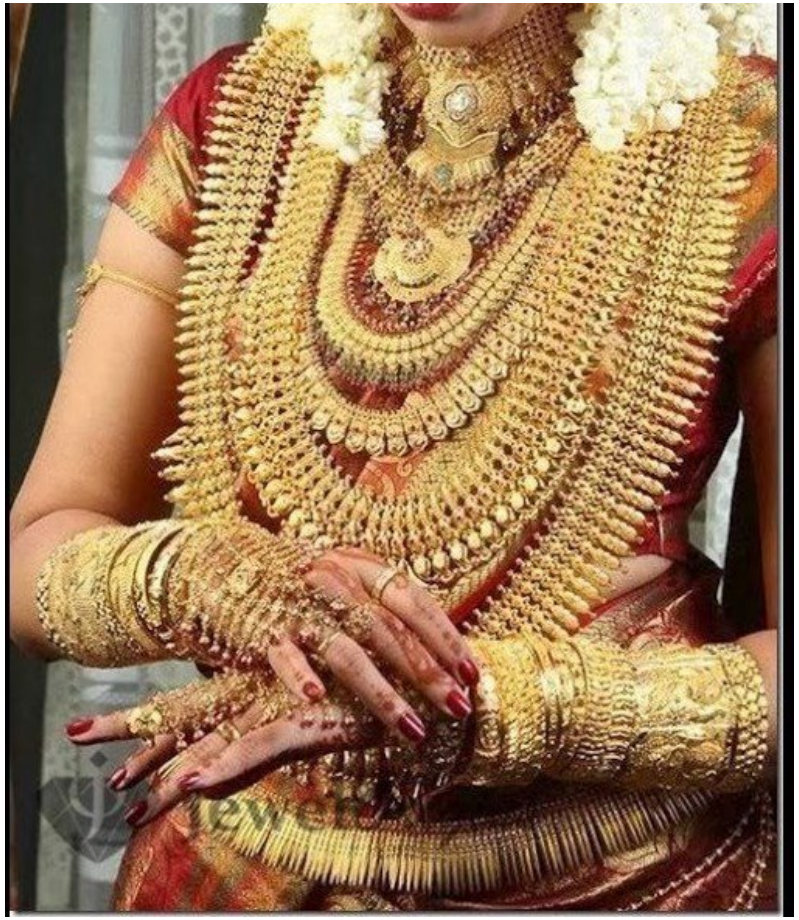


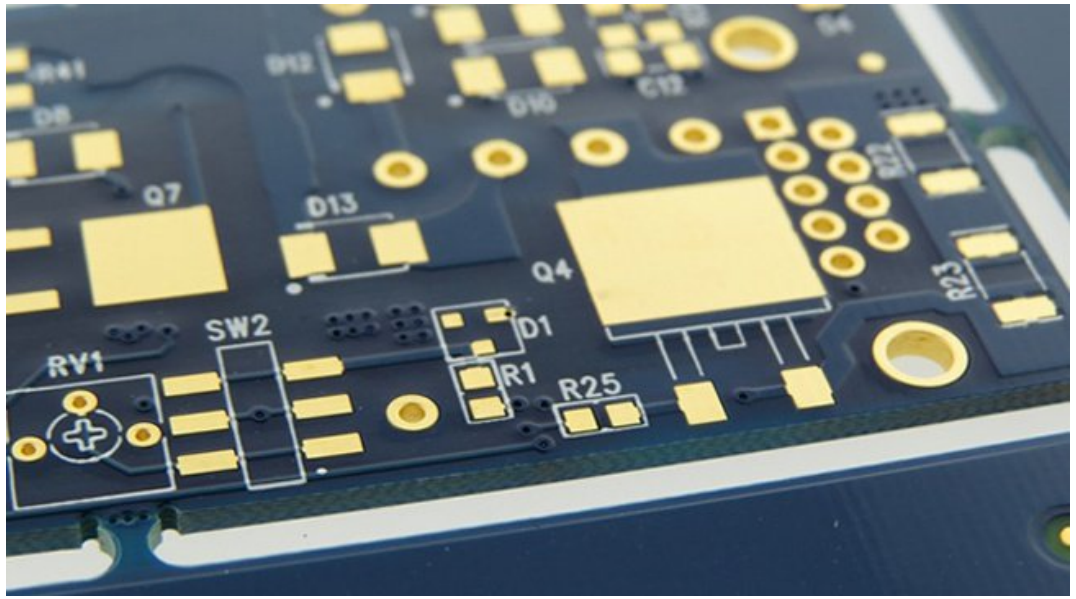




- compatible with human body
- Flexible
- biodegradable

gold



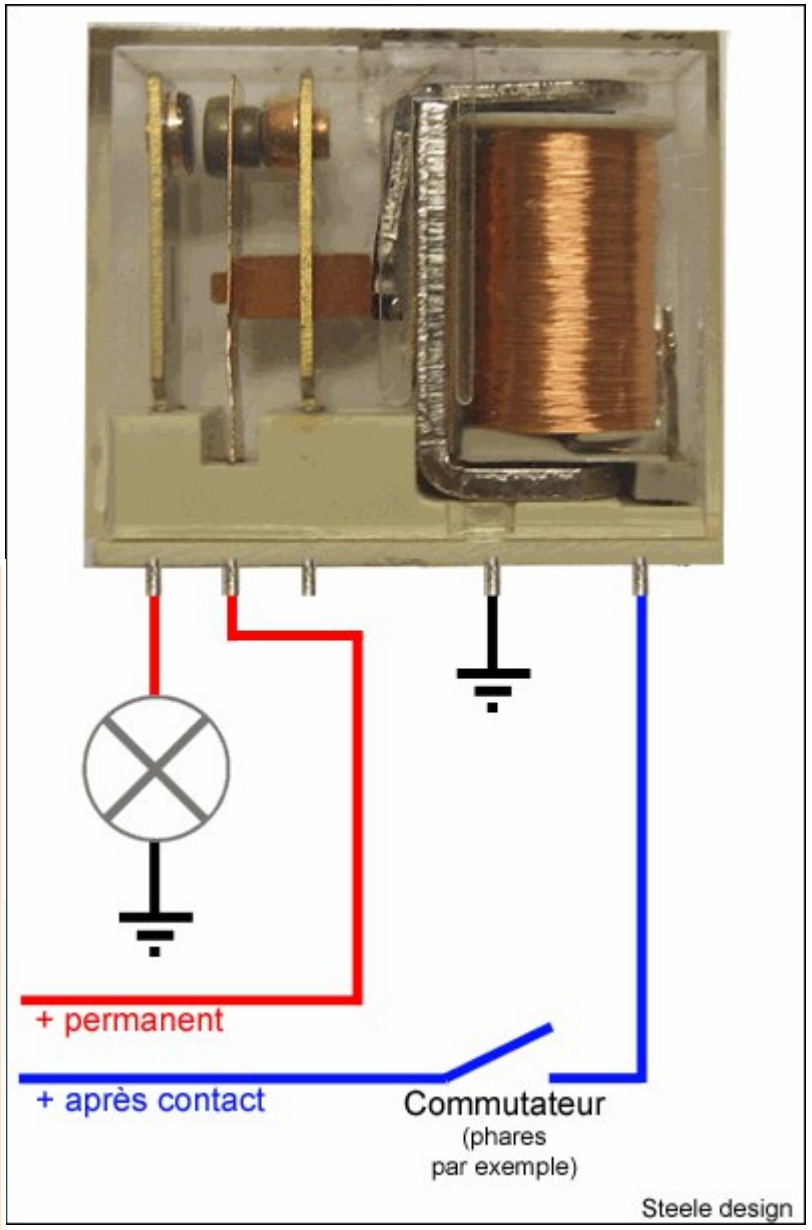


- Inert
- Good electrical conductor
- Ductile and malleable
- expensive

silver




YU
ZHANG
ISO9001:2008
www.shyzrivet.com


+ permanent
+ après contact
Commutateur
(phares
par exemple)
Steele design

- Good electrical conductor
- Prone to oxidation
- Good thermal and electrical conductor
- expensive

diamond







- Hard
- Good thermal conductor
- brittle

bamboo

• .











- Strong
- Biodegradable
- Prone to insect attack

paper










Egg Printing Explained
Farming Method
0 = Organic
1 = Free Range
2 = Barn
3 = Cage
XX = Country of Origin
E.g. FR = Fran
0XX ABC12
Farm ID
EGGS LAID BY HENS ON A VEGETARIAN ORGANIC DIET AND FREE TO ROAM ON ORGANIC PASTURE DURING DAYLIGHT HOURS





- Biodegradable
- Flexible
- combustible

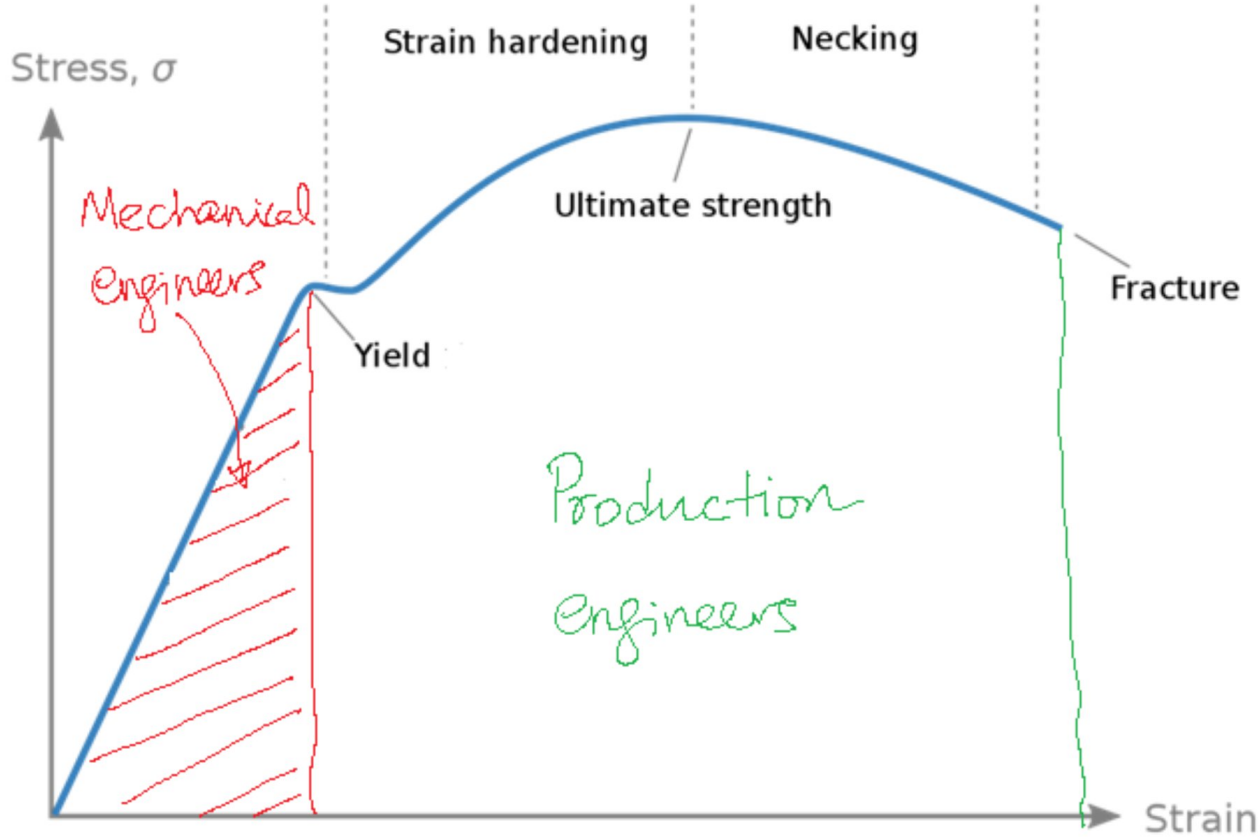
Properties of materials

- 1. stiffness: deflection
- 2. strength: tension, compression, shear
- 3. hardness
- 4. corrosion resistance
- 5. density
- 6. toughness
- 7. ductility
- 8. malleability
- 9. thermal conductivity
- 10 electrical conductivity
- 11.magnetic property
- 12. recyclability
- 13. availability
- 14. safety
- 15 combustibility
- 16. cost



Mechanical Vs production engineers domain of operation in stress –strain curve

- Mechanical engineers design components which should not deform plastically during its intended operation. They work in elastic region of materials
- Production engineers needs to apply stresses to plastically deform materials to change its shape but it should not break. They operate in plastic region of the stress- strain diagram



Ductility is a solid material's ability to deform under tensile stress; this is often characterized by the material's ability to be stretched into a wire.

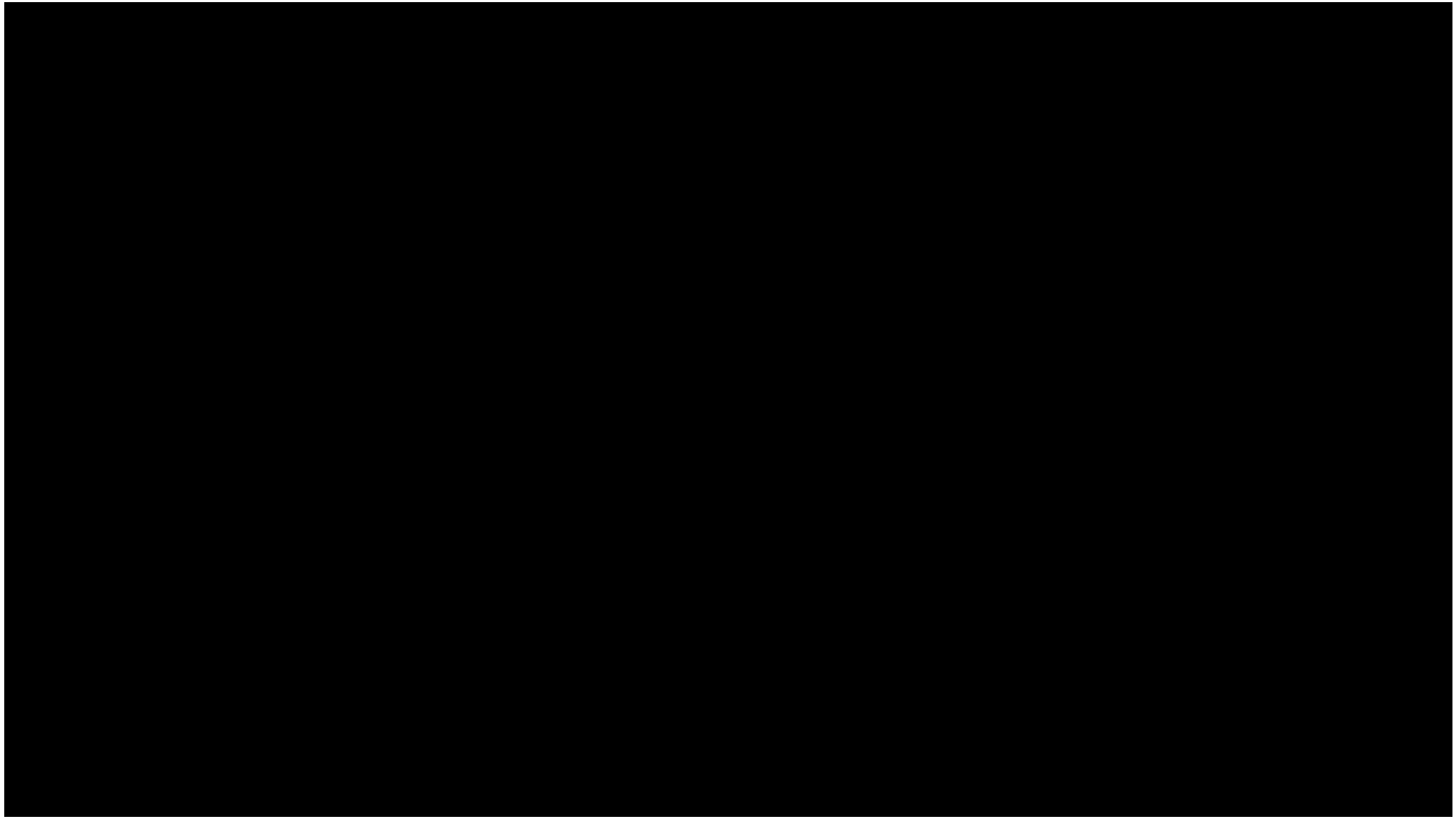
Malleability, is a material's ability to deform under compressive stress without fracture; material's ability to be deformed into a thin sheet by hammering or rolling .

Toughness

- Toughness is the ability of a material to absorb energy and plastically deform without fracturing. It may be presented as impact strength of material.
- The environment in which a material is used can affect its toughness. Exposure to corrosive environments, high temperatures, and radiation can reduce a material's toughness.
- Toughness can also refer to the quality of being strong and not easily broken or damaged. For example, you might describe someone as having toughness and resilience.

hardness

- Defined as resistance to permanent indentation
- Also defined as resistance to scratching or wear
- Hardest material is diamond
- Hardness scales; Brilnell, Vickers, Knoop, Rockwell



Hardenability is the ability of material to get hardened by hardening process.

Stiffness is the rigidity of an object i.e. the extent to which it resists deformation in response to an applied force.
Force applied but no deformation = Very Stiff.

Toughness is the ability of a material to absorb energy and plastically deform without fracturing.
Force applied, plastically deformed but no fracture = Very tough.